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DOES THE NUMBER OF HARVESTED SENTINEL LYMPH NODES MATTER IN BREAST CANCER PATIENTS WITH LIMITED NODAL BURDEN? A PROSPECTIVE STUDY

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BACKGROUND

In breast cancer patients with limited nodal burden, sentinel lymph node biopsy (SLNB) alone, without axillary lymph node dissection (ALND), has non-inferior survival[1]. However, it remains unclear if the number of SLNs will affect the accuracy of axillary staging and oncological outcomes.

OBJECTIVES

We aimed to determine if the number of harvested SLNs affects oncological outcomes.

METHODS

In this prospective study, T1-2 breast cancer patients with 1-2 metastatic SLNs and no ALND were enrolled from KK Women's and Children's Hospital, Singapore, from Jan 2020 to Jan 2026. Patients' demographics and pathological data were analyzed against patients who underwent ALND.
Categorical and continuous variables were analyzed using Fisher's exact test and t-test respectively. P-value < 0.05 was considered statistically significant. Statistical analysis was done using Graphpad statistical software (version 2025).

RESULTS

152 patients were enrolled, of which 63 and 89 underwent SLNB alone and ALND respectively. The 2 groups were comparable except that ALND group had more patients with mastectomy (p=0.0044), nodal radiation (p=0.0277) and macrometastasis (p=0.0001) (Table 1).

Median number of harvested SLNs was 3(1-8) and 2(1-9) in the SLNB and ALND groups respectively. In the ALND group, residual median metastatic nodes of 1.5 (1-9) was seen in 24.7%.

Harvested SLNs of 1 (p=0.7551), 1-2 (p=0.2263) or 1-3 (p=1.0000) did not detect more metastatic non-SLNs.

Single agent SLNB localization was used in 22% and 19.3% of SLNB and ALND group respectively. Though dual agent localization had more patients with >/=2 SLNs (p=0.0181), it did not detect more metastatic non-SLNs (P=0.1184) compared to single agent localization.

Clinical features	No. (%) of patients with SLNB alone N=63	No. (%) of patients with ALND N=89	P -Value
Age (years)			0.8672
≤50	26 (41%)	35 (39%)	
>50	37 (59%)	54 (61%)	
Pathological features			
Breast tumour histology			0.4910
Invasive ductal	49 (78%)	68 (76%)	
Invasive lobular	5 (8%)	12 (13%)	
Others	9 (14%)	9 (10%)	
Grade			0.2425
I	4 (6%)	11 (12%)	
II	47 (75%)	55 (62%)	
III	12 (19%)	23 (26%)	
Estrogen receptor (ER)			0.7418
positive	58 (92%)	84 (94.4%)	
negative	5 (8%)	5 (5.6%)	
Progesterone receptor (PR)			0.1148
positive	57 (90%)	72 (81%)	
negative	6 (10%)	17 (19%)	
Human epidermal growth factor receptor 2 (HER2)			0.8220
Positive	9 (14%)	15 (17%)	
negative	54 (86%)	74 (83%)	
pT			1.0000
T1	27 (43%)	37 (42%)	
T2	36 (57%)	52 (58%)	
Number of SLN harvested			0.2007
1	6 (10%)	17 (19%)	
2	19 (30%)	29 (33%)	
>2	38 (60%)	43 (48%)	
Number of metastatic lymph nodes on ALND			NA
0	NA	67 (75.3%)	
1	NA	11 (12.3%)	
>1	NA	11 (12.3%)	
Lymph node metastasis			0.0001
Micrometastasis	38 (60.3)	8 (9.0)	
Macrometastasis	25 (39.7)	81 (91.0)	
Surgical management of the Breast			0.0044
Mastectomy	30 (48%)	63 (71%)	
Breast conservation	33 (52%)	26 (29%)	
Adjuvant Radiotherapy			0.0760
Yes	55 (90%)	70 (79%)	
No	6 (10%)	19 (21%)	
Missing	2	0	
Nodal Irradiation			0.0277
Yes	35 (59%)	67 (77%)	
No	24 (41%)	20 (23%)	
Missing	4	2	
Recurrence			1.0000
Yes	0 (0)	0 (0)	
No	63 (100%)	89 (100%)	

TABLE 1. Characteristics of patients who underwent sentinel lymph node biopsy (SLNB) alone versus axillary lymph node dissection (ALND).

In the SLNB group, 11.1% had 1/1 or 2/2 metastatic nodes. There was no recurrence after a median follow-up of 37 (3-75) months.

Discussion

Our paper demonstrated that the number of harvested SLNs did not affect oncological safety. This is consistent with the results from another study which demonstrated that omission of ALND in patients with less than 3 SLNs did not increase the risk of disease recurrence[2].

Strengths of this paper include it being one of the few papers in literature which specifically studied the relationship of the number of harvested SLNs with oncological outcomes in patients with limited nodal burden. This has practical implications as it could guide the number of SLNs to be harvested and avoid arm lymphedema[3] due to excessive SLN harvest. SLNB alone may also be possible in patients with up to 2 SLNs harvested which were metastatic.

CONCLUSION / TAKE-HOME MESSAGE

The number of harvested SLNs and the use of a single localization agent did not affect oncological outcomes in patients with limited nodal burden. Harvesting additional non-SLNs was unnecessary.

Omission of ALND may be a safe approach even in patients with 1/1 or 2/2 metastatic SLNs, with no increased recurrence, based on medium term follow-up.

Our findings need validation in larger studies with longer follow-up.

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This study was registered with ClinicalTrials.gov (identifier: NCT06259513).

Conflict of interest: nil